

Mingyu Wang

New Taipei City, Taiwan

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Data Analyst | Financial & Risk Analytics | Python / SQL / R

PROFESSIONAL SUMMARY

Data Analyst with a background in Information Management and postgraduate training in Financial Technology with Data Science at the University of Bristol. Experienced in Python, SQL, R, large-scale data analysis, financial and transaction data, customer analytics, statistical modelling, relational data modelling, and reproducible analytical workflows. Skilled at turning complex or loosely defined datasets into structured, interpretable analysis to support business, risk, and product decision-making.

CORE SKILLS

Analytics and Programming: Python, SQL, R, pandas, NumPy, Polars, scikit-learn, exploratory data analysis, statistical modelling, feature engineering, segmentation, data visualisation

Financial and Risk Analytics: financial market data, transaction data, customer behaviour analytics, risk indicators, market microstructure, high-frequency data, order-book analytics

Data Systems and Validation: PostgreSQL, Supabase, relational data modelling, data cleaning, data validation, data quality checks, reproducible pipelines, cross-validation, holdout testing

Reporting and Tools: Tableau, Jupyter Notebook, R Markdown, Git, GitHub, AWS (EC2, S3, SQS, DynamoDB, CloudWatch)

EDUCATION

University of Bristol

MSc Financial Technology with Data Science

Bristol, United Kingdom

Sep 2025 – Sep 2026

- Global rankings: TIME World's Top Universities 2026 #48; QS World University Rankings 2027 #57; Times Higher Education World University Rankings 2026 =#80.
- Completed all taught modules, MSc Individual Project, and viva examination; final programme results and formal degree award pending, with results expected in November 2026.

National Kaohsiung University of Science and Technology

Bachelor of Science in Information Management

Kaohsiung, Taiwan

Sep 2020 – Jun 2024

- Studied data science, database systems, software engineering, big data analytics, social network analysis, and information systems, with projects using Python, R, SQL, databases, data processing, and visualisation.

SELECTED ANALYTICS PROJECTS

High-Frequency Liquidity Shock Detection and Order-Book Resiliency | *MSc Individual Project*

- Analysed more than 8 million rows of high-frequency Nasdaq order-book data across 10 securities and 37 trading sessions to detect liquidity shocks and measure market recovery.
- Built an outcome-blind event-detection and recovery framework using time-of-day median/MAD benchmarks, first-passage T50/T90 measures, censor-aware analysis, and systematic pipeline validation.
- Applied Kaplan–Meier estimation and security-stratified log-rank testing to compare spread and depth recovery, identifying materially slower recovery in displayed depth.
- Validated 34,589 event records across 3.73 million event-second recovery rows with zero dropped or duplicated events and independently checked recovery calculations.

Customer Behaviour Risk Analytics & Segmentation | *R / Customer Analytics*

- Analysed trading data from 1,000 customers and transformed transaction-level activity into interpretable customer-level behavioural and risk indicators.
- Engineered signals for high-exposure trades, FOMO-like buying, market-order reliance, and repeated deposits under low-balance conditions.
- Identified the 200 most active customers and selected a 40-customer target cohort for behavioural risk segmentation and prioritised monitoring.
- Implemented the analysis in a reproducible R Markdown workflow covering feature engineering, aggregation, cohort selection, visualisation, and documented limitations.

MindPass FinTech Escrow DApp | *FinTech / Relational Data*

- Developed a four-person FinTech MVP combining smart-contract escrow with a PostgreSQL / Supabase relational data layer for booking, funding, refund, subsidy, and settlement workflows.
- Designed SQL-backed relational structures connecting users, sessions, funding records, transaction records, subsidy state, and settlement workflows.
- Integrated on-chain events with application data to maintain consistent booking, funding, refund, and settlement states across the workflow.
- Led technical implementation, integration, and validation across funding, provider decisions, check-in, refunds, balances, and withdrawal processes.

Robust Optimisation and Predictive Modelling Pipeline | *R / Statistical Modelling*

- Built reproducible modelling workflows covering exploratory analysis, model comparison, validation, holdout evaluation, and interpretation.
- Evaluated five Differential Evolution configurations across 26 benchmark problems using blocked experimental designs to assess robustness and decision stability.
- Built a leakage-aware predictive modelling workflow using group-aware data splitting, grouped 5-fold cross-validation, and held-out evaluation.

LEADERSHIP AND ADDITIONAL INFORMATION

Leadership: Chair, Student Judicial Council; Deputy Speaker, Student Parliament, with experience in structured review, documentation, meeting coordination, and stakeholder communication.

Languages: Mandarin Chinese: Native; English: Professional working proficiency; MSc programme completed in English.

Availability: Based in New Taipei City; available for full-time roles from October 2026. **Work Authorisation:** Eligible to work in Taiwan.